Learning to Ask Informative Questions: Enhancing LLMs with Preference Optimization and Expected Information Gain

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Abstract

Questions are essential tools for acquiring the necessary information to complete information-seeking tasks. However, large language models (LLMs), especially open-source models, often perform poorly in generating informative questions, as measured by expected information gain (EIG). In this paper, we propose a method to enhance the informativeness of LLM-generated questions in 20-question game dialogues. We sample multiple questions from the same model (LLAMA 2-CHAT 7B) for each game and create pairs of low-EIG and high-EIG questions to apply a Direct Preference Optimization (DPO) algorithm. Our results show that this method produces more effective questions (in terms of EIG), even in domains different from those used to train the DPO model.

1 Introduction

Questions in language serve as requests for information (Searle, 1969). A speaker lacks information in their knowledge state and asks questions to gain this information. This process of acquiring information through questioning is essential for children to learn about the world (Chouinard, 2007) and for adults to solve complex problems (Geva et al., 2021). Questions, however, vary in their level of informativeness, with some questions being more informative and efficient in reaching the problem's solution (Grand et al., 2023).

Cognitive Science provides two interesting tools to study questions' informativeness: the 20 Questions Game as a test bed, and the Expected Information Gain (EIG) as a quantifying measure (Ruggeri and Lombrozo, 2015). The 20 Questions game consists of one player asking yes/no questions to identify the item the other player has in mind, from a pool of possible items. In this context, the EIG (Shannon, 1948) measures questions' informativeness as the (expected) entropy reduction caused by a certain question in the space of possible items. Most informative polar questions partition the possible items into two same-size clusters: items in one cluster are expected to receive a positive answer, and items in the other cluster are expected to receive a negative one.

In recent years, Large Language Models (LLMs) have demonstrated remarkable language and reasoning capabilities (Kojima et al., 2022; Huang and Chang, 2023). The ability to ask informative and effective questions is crucial for employing these systems as successful user assistants on a large scale. Although LLMs have demonstrated their ability to play the 20 Questions game, their questions are characterised by low informativeness and limited success (Bertolazzi et al., 2023). Recent studies have emerged to improve the informativeness of LLMs' questions. Zhang et al. (2024) improve open-source LLMs via Reinforcement Learning and Behavioral Cloning from larger LLMs' data. Relying on LLMs' ability to generate diverse questions and provide reliable answers (Piriyakulkij et al., 2023; Testoni et al., 2023), Hu et al. (2024) propose an inference-time re-ranking strategy based on EIG. Differently from them, we propose a training strategy leveraging EIG as a refined signal with a preference optimization algorithm. Unlike other approaches, all the steps of our method are accomplished by the same open-source model, without requiring annotation or feedback signals from external models.

This study aims to improve the informativeness of questions generated by LLMs, thereby leading to more efficient agents. The 20 Questions Game serves as a testbed to illustrate the feasibility of our approach. To achieve our aim, we propose a method involving three steps: sampling multiple questions, questions' evaluation in terms of EIG, and training with preference optimization. In the first step, the model generates a set of possible questions, and it computes the EIG for each question. This set of questions, along with the corresponding

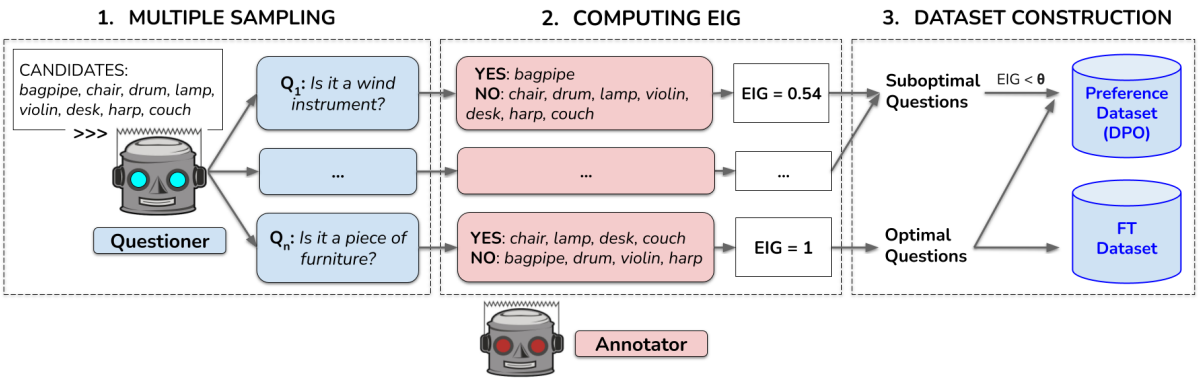

Figure 1: The proposed approach for constructing the datasets of dialogues for fine-tuning and preference optimization (DPO). From the original candidate set, the Questioner generates a question Q_1, and the Annotator provides the expected answer to each candidate. Expected Information Gain (EIG) is computed from the annotation: if the question is suboptimal in terms of EIG, other questions are sampled until an optimal question is reached (Q_n). The optimal question is paired with the suboptimal ones in the Preference dataset (DPO), whereas the Fine-Tuning (FT) dataset is composed of only 1-EIG questions.

EIG values, as a proxy for the questions' informativeness, is used for Direct Preference Optimization (DPO; Rafailov et al. 2023). Our results show that EIG is a strong training signal to improve the question-asking capabilities of current LLMs and overcome their shortcomings in asking effective questions.[1]

While these findings hold in the controlled setting of the 20 Questions Game, our approach could be extended outside this paradigm and metric to train different models' capabilities. Our core idea of sampling, self-annotation to filter with a metric, and preference optimization could improve questions' generations in a plethora of tasks – recommendation systems (Piriyakulkij et al., 2023) and image retrieval (Keh et al., 2023) for example. Other metrics, such as Expected Savings (Rothe et al., 2018), could be employed to generate large preference datasets and improve LLMs' information-seeking abilities beyond our tested paradigm.

2 Methodology

2.1 Setting

Our experimental setting, inspired by the 20 Questions Game paradigm, involves two players with different roles: a Questioner and an Answer. The Answer is secretly assigned one target entity ω among a pre-defined set of candidates (the *candidate set* – Ω). The Questioner receives the initial candidates and is instructed to ask yes/no questions, in order to identify ω.

Each *game* consists of a set of possible items, Ω, available to the Questioner, and a specific target item, ω, available to the Answerer, where $\omega \in \Omega$. A *dialogue* is a series of question-answer exchanges between the two players, ending when the Questioner correctly identifies the target or reaches the maximum number of questions (20). If the target is reached within the 20 questions, the dialogue is *successful*; otherwise, it is *unsuccessful* (see more details on the successful condition in Appendix A.1). For a given game, if the dialogue is not successful on the first attempt, the Questioner and the Answerer can engage in up to n dialogues to identify the target, where $n = |\Omega|/2$.

2.2 Data Sampling

Following the procedure in Bertolazzi et al. (2023), 10,000 games are extracted from McRae concepts (McRae et al., 2005). The candidate sets consist of concepts belonging to 6 categories (mammal, bird, clothing, weapon, fruit, and vegetable) for a total of 66 unique concepts. Each candidate set consists of 8 elements ($|\Omega| = 8$) belonging to two categories of 4 elements each.

In our setting, all the game's roles are played by the same LLM, LLAMA 2-CHAT 7B (Touvron et al., 2023). Along with the Questioner and the Answerer, sampling involves the Annotator, who provides the annotation to compute the EIG. As illustrated in Fig. 1, English dialogues are generated from each game by repeatedly sampling questions

[1]Code and data are available at https://github.com/dmazzaccara/LearningToAsk

from the Questioner. For each generated question, the Annotator provides the yes/no annotation (Ω_{yes} and Ω_{no}). If the question splits the candidate sets in two equal subsets ($|\Omega_{yes}| = |\Omega_{no}|$), the question is *optimal* in terms of EIG (details on EIG computation are in Appendix A.2). Once an optimal question is reached, it is saved along with the answer by the Answerer in the dialogue history. This dialogue history is provided as input to the Questioner to sample the optimal follow-up question (implementation's details are in Appendix A.3).

This process results in two clusters of optimal and suboptimal questions, from which two datasets are obtained. A *fine-tuning dataset* consisting of dialogues of only optimal questions ($EIG = 1$), indicating that they evenly split the candidate space at each turn. A *preference dataset*, which consists of 55,000 optimal ($EIG = 1$) vs suboptimal ($EIG < 0.8$) question pairs.

2.3 Questioner Models

Our evaluation of the Questioner compares LLAMA 2-CHAT (7B) zero-shot with the trained versions below. Both the training processes rely on LoRA adapters and standard hyperparameter settings.

Fine-tuning (FT) involves a causal language modelling objective applied to the entire sequence. For development, we use 50 games excluded from the fine-tuning dataset. To obtain the best adapter's checkpoint, we test the adapter on the development set every 1,000 training samples. According to the number of questions per dialogue, the best adapter's checkpoints are after 4,000 dialogues.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) training objective increases the likelihood of optimal EIG questions while decreasing the likelihood of the low-EIG question. For DPO, we train with all the 55,000 pairs of optimal vs low-EIG questions.

2.4 Evaluation

Test Sets: three test sets evaluate questioner models with candidate sets of the same size as in training ($|\Omega| = 8$): INLG, Things, Celebrities; two test sets test with more candidates than in training: INLG 16 and BigBench. The same-size test sets have candidate sets from different domains: INLG (Bertolazzi et al., 2023) from seen categories but unseen concepts; both from Zhang et al. (2024), Things has unseen categories but common-life concepts, Celebrities has unseen categories and concepts compared to training (famous personalities). Different-size test sets are INLG 16 (Bertolazzi et al., 2023), with sets of $|\Omega| = 16$ unseen concepts of seen categories, and BigBench, with sets of $|\Omega| = 29$ different categories and concepts (Srivastava and al., 2023). Details are in Appendix B.

Metrics: the impact of FT and DPO on zero-shot is assessed with three metrics. Task success (**S@1**) is the percentage of games in which the model identifies the target within the first dialogue. Average number of questions (**AQ**) is the number of questions to reach the target in successful dialogues. Expected Information Gain (**EIG**) is the averaged questions' EIG in successful dialogues.

3 Results

Different methods are assessed on test sets with candidate sets of the same size as in training but from other domains. The results are reported in Table 1. For INLG and Things, DPO significantly improves performance over the zero-shot model in terms of both S@1 ($+12.2\%$ and $+10\%$) and AQ (-2.1 and -2.3 questions per successful dialogues). For Celebrities, where the concept space differs greatly from the training data, DPO reduces the average number of questions by 2.5, while S@1 shows only a marginal improvement. A consistent improvement of DPO over zero-shot is also observed for the average EIG of the generated questions. Overall, the fine-tuning approach leads to a significant degradation in the S@1 metric.

Set	Method	S@1 ↑	AQ ↓	EIG ↑
INLG	zero-shot	56.7%	7.1	0.34
	FT	46.6%	**4.6**	0.41
	DPO	**68.9%**	5.2	**0.45**
Things	zero-shot	51.1%	7.5	0.29
	FT	42.2%	5.4	0.31
	DPO	**61.1%**	**5.2**	**0.40**
Celebrities	zero-shot	71.1%	7.6	0.35
	FT	46.7%	5.5	0.39
	DPO	72.2%	**5.1**	**0.47**

Table 1: **Different Domains**: Results on candidate sets with $|\Omega| = 8$. Across the three sets, DPO identifies the target within the first dialogue (S@1) more often, with shorter dialogues (AQ) and more informative questions (EIG). FT and DPO's improvements for AQ and EIG are statistically significant compared to zero-shot scores (Mann-Whitney U test).

Given the promising results of DPO, we proceed to evaluate the robustness of this approach with